

ALGORITHMIC NON-COMMUTATIVE CLASS FIELD THEORY, XV: THE PARAHORIC FACTORIZATION OF THE GEODESIC EDGE FLOW, THE UNITARITY OF THE REMAINDER, AND THE CHAMBER COMPANION

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ABSTRACT. [XI, Cor. 3.5] found numerically, on one $\tilde{A}_2$ complex, that the characteristic polynomial of the geodesic edge flow L_E factors as $\det(I - uL_E)(1 - u^3)(1 - q^3u^3) = \det P(u)G(u)$, with P the cubic Hecke pencil and G a polynomial of degree $4n + 6$ all of whose roots lie on $|u| = q^{-1/2}$, and left the proof and the identification of G open. This paper proves the identity for every $\tilde{A}_2$ complex of order q , identifies G , and proves the Riemann hypothesis for it unconditionally. The mechanism is a two-sided intertwining: the image of $\Psi = [S^t T^t M^t]$ is invariant under L_E and L_E^t , so its orthogonal complement $W = \ker S \cap \ker T \cap \ker M$ reduces L_E , and the local identities $L_E L_E^t = qI + (q^2 - q)T^t T$, $L_E^t L_E = qI + (q^2 - q)S^t S$ — two points of a projective plane lie on one line — make $R := L_E|_W$ satisfy $RR^t = R^t R = qI$. Hence $R/\sqrt{q}$ is orthogonal, $G(u) = \det(I - uR)$, and every root of G has $|u| = q^{-1/2}$ for every complex, Ramanujan or not; the Ramanujan property of the complex lives entirely in the vertex pencil. The same calculus on pointed chambers gives the Kang–Li operator L_B a companion B on $\ell^2(E)^3$, with $L_B \Phi = \Phi B$ and $\det(I + uB) = \det((1 + qu^3)I + uK + u^2 K^t) = (1 - u^3)^n \det(I - uL_E) \det(I - u^2 L_E) / \det P(u)$: the Kang–Li zeta identity in companion form, with the parahoric block $G(u)G(u^2)$ explaining the eigenvalues of modulus $q^{1/4}$ of the chamber operator as square roots of the unitary remainder. A correction to [XI, Def. 3.1] is recorded (the geodesic condition is “not in a common chamber”). Verified exactly on $Y_{21}, Y_{24}, Y_{42}, T_6$ and structurally on Y_{168} .

STATUS OF THE RESULTS

Every numbered statement is proved; the proofs use only the link structure of an $\tilde{A}_2$ complex of order q ([X, Def. 1.1(i)–(iii)], vertex links the incidence graph of $\text{PG}(2, q)$), linear algebra over $\mathbb{Q}$, and the results of [XI], [XII] that are re-derived where used. Kang–Li [1] and Kang–Li–Wang [2] are cited for context and for the one statement we do not reprove (the relation between L_B and its companion for $q \neq 2$, Corollary 4.3); Lubotzky–Samuels–Vishne [3] for the notion of Ramanujan complex. The flag condition [X, Def. 1.1(iv)] is not assumed anywhere.

1. SETTING, AND A CORRECTION

Y is a finite $\tilde{A}_2$ complex of order q : a connected pure 2-dimensional simplicial complex with a $\mathbb{Z}/3$ vertex typing, n vertices, typed edges E ($e = (x \rightarrow y)$, $\tau(y) = \tau(x) + 1$, $|E| = n(q^2 + q + 1) =: nN$), chambers F (one vertex of each type, $|F| = (q+1)|E|/3$), such that the link of every vertex y is the incidence graph of $\text{PG}(2, q)$: in-neighbours of y are the points, out-neighbours the lines, and $\{x, y, z\}$ is a chamber iff the point x lies on the line z . $A_1, A_2 = A_1^t$ are the Hecke operators, $P(u) := I - A_1 u + q A_2 u^2 - q^3 u^3$ the pencil, $S, T, M \in M_{n \times |E|}(\mathbb{Z})$ the tail, head and third-vertex

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Paper XV of the series; Papers I–XIV are cited as [I]–[XIV]. The verification script `pgl3_parahoric.py` in the ancillary bundle reproduces every computational claim; its output is archived as `pgl3_parahoric_output.txt`.

incidences, $K := \sum_i d_i d_{i+1}^t$ the chamber continuation ($K(e, e') = \#\{\text{chambers in which } e' \text{ follows } e\}$), all as in [XI, §3] and [XII, Thm. 3.1].

Definition 1.1 (Geodesic edge flow, chamber form). $L_E \in M_{|E|}(\{0, 1\})$, $L_E((x \rightarrow y), (y \rightarrow z)) = 1$ iff $(y \rightarrow z) \in E$ and $\{x, y, z\} \notin F$; equivalently $L_E = T^t S - K$ [XII, Thm. 3.1].

Remark 1.2 (Correction to [XI, Def. 3.1]). [XI, Def. 3.1] used “ z not adjacent to x ”. The two conditions agree exactly when Y is a flag complex ([X, Def. 1.1(iv)]), which holds for Y_{168} , the only complex on which [XI] evaluated L_E . On Y_{21} , whose 1-skeleton is $K_{7,7,7}$, every z is adjacent to x and the adjacency version is the zero matrix, while Definition 1.1 gives row sums q^2 ; on Y_{24} and Y_{42} the two versions differ as well (Table 1, line (D)). Definition 1.1 is the projection of the geodesic operator of the building, the operator L_E of [1], and every statement of [XI, §3] and [XII] holds for it with the proofs unchanged; the quoted numerics of [XI] concern Y_{168} and are unaffected.

Lemma 1.3 (Local counts). For $e = (x \rightarrow y)$, $e' = (x' \rightarrow y)$ with the same head, and $f \in \ell^2(V)$, $m_f := M^t f$:

- (a) e has q^2 geodesic continuations and q^2 geodesic predecessors;
- (b) $\sum_{z: e \rightarrow (y \rightarrow z)} f(z) = (A_1 f)(y) - m_f(e)$, $\sum_z m_f(y \rightarrow z) = q((A_2 f)(y) - f(x))$;
- (c) $\#\{\text{common geodesic continuations of } e, e'\} = q^2$ if $x = x'$ and $q^2 - q$ if $x \neq x'$; dually for two edges with the same tail;
- (d) $\sum_{x: (x \rightarrow y) \rightarrow e''} f(x) = (A_2 f)(y) - m_f(e'')$ and $\sum_x m_f(x \rightarrow y) = q((A_1 f)(y) - f(z))$ for $e'' = (y \rightarrow z)$.

Proof. In the link of y the continuations of e are the lines z not through the point x : $N - (q+1) = q^2$. (b) is [XI, Lem. 3.2(b),(c)] with “chamber” in place of “adjacent”, the proof verbatim. (c): lines through neither of two distinct points: $N - 2(q+1) + 1 = q^2 - q$. (d) is (b) for the opposite complex (all edges reversed, types negated), whose L_E is L_E^t . $\square$

2. TWO-SIDED INTERTWINING AND THE UNITARY REMAINDER

Let $\Psi := [S^t \ T^t \ M^t]: \ell^2(V)^3 \rightarrow \ell^2(E)$ and $W := (\text{im } \Psi)^\perp = \ker S \cap \ker T \cap \ker M$.

Theorem 2.1 (Two-sided intertwining). With $I = I_n$ and block matrices over $\ell^2(V)^3$,

$$\begin{aligned} L_E \Psi &= \Psi C', \quad C' = \begin{pmatrix} 0 & 0 & -qI \\ q^2 I & A_1 & qA_2 \\ 0 & -I & 0 \end{pmatrix}; & L_E^t \Psi &= \Psi C'', \quad C'' = \begin{pmatrix} A_2 & q^2 I & qA_1 \\ 0 & 0 & -qI \\ -I & 0 & 0 \end{pmatrix}; \\ K \Psi &= \Psi \hat{C}, \quad \hat{C} = \begin{pmatrix} 0 & 0 & qI \\ (q+1)I & 0 & A_2 \\ 0 & I & 0 \end{pmatrix}; & K^t \Psi &= \Psi \hat{C}', \quad \hat{C}' = \begin{pmatrix} 0 & (q+1)I & A_1 \\ 0 & 0 & qI \\ I & 0 & 0 \end{pmatrix}; \\ L_E L_E^t &= qI_E + (q^2 - q)T^t T, & L_E^t L_E &= qI_E + (q^2 - q)S^t S. \end{aligned}$$

Consequently $\text{im } \Psi$ and W are invariant under L_E, L_E^t, K, K^t , and $\ell^2(E) = \text{im } \Psi \oplus W$ is an orthogonal decomposition reducing all four.

Proof. The first identity is [XI, Thm. 3.3], whose proof uses only Lemma 1.3(a),(b). For L_E^t : $(L_E^t S^t f)(y \rightarrow z) = \sum_x f(x)$ over geodesic predecessors, which is $(A_2 f)(y) - m_f(y \rightarrow z)$ by (d), i.e. $L_E^t S^t = S^t A_2 - M^t$; $(L_E^t T^t f)(y \rightarrow z) = q^2 f(y)$ by (a), i.e. $L_E^t T^t = q^2 S^t$; $(L_E^t M^t f)(y \rightarrow z) = \sum_x m_f(x \rightarrow y) = q((A_1 f)(y) - f(z))$ by (d), i.e. $L_E^t M^t = qS^t A_1 - qT^t$. Reading the three columns gives C'' . For $K = T^t S - L_E$: $T^t S \Psi = T^t [SS^t \ ST^t \ SM^t] = T^t [NI \ A_1 \ (q+1)A_2]$ by [XII, Thm. 3.1], and subtracting $L_E \Psi = \Psi C'$ gives $K S^t = (q+1)T^t$, $K T^t = M^t$, $K M^t = T^t A_2 + qS^t$, i.e. $\hat{C}$; $K^t = S^t T - L_E^t$ likewise gives $\hat{C}'$. The two product identities are Lemma 1.3(c): $(L_E L_E^t)(e, e')$ counts common continuations, zero unless the heads agree, q^2 on the diagonal and $q^2 - q$ off it, and

$(T^t T)(e, e') = [\text{head}(e) = \text{head}(e')]$. Invariance of $\text{im } \Psi$ under an operator X and of $W = (\text{im } \Psi)^\perp$ under X^t are the same statement; since all four operators and their transposes preserve $\text{im } \Psi$, both summands are invariant under all four. Verified as exact integer identities on all five complexes (Table 1, line (D)). $\square$

Theorem 2.2 (The remainder is $\sqrt{q}$ times an orthogonal matrix). *Let $R := L_E|_W$. Then $R^t = L_E^t|_W$ and*

$$RR^t = R^t R = q I_W.$$

Hence R is invertible, every eigenvalue of R has modulus $\sqrt{q}$, and $G(u) := \det(I - uR) \in \mathbb{Z}[u]$ has all its roots on the circle $|u| = q^{-1/2}$, with $G(u) = (-u)^d \det(R) G(1/(qu))$, $d = \dim W$, $\det R = \pm q^{d/2}$.

Proof. W is invariant under L_E and L_E^t (Theorem 2.1), so the restriction of L_E^t to W is the adjoint of R . For $v \in W$, $Tv = 0$ and $Sv = 0$, so the product identities give $L_E L_E^t v = qv$ and $L_E^t L_E v = qv$. Thus $R/\sqrt{q}$ is orthogonal, its eigenvalues have modulus one, and $\det(I - uR) = \det(-uR) \det(I - (uR)^{-1}) = (-u)^d \det R \cdot \det(I - \frac{1}{qu} R^t)$. $G \in \mathbb{Z}[u]$ because $W \cap \mathbb{Z}^E$ is a saturated L_E -invariant sublattice of full rank in W . On the five complexes the eigenvalues of R have modulus $\sqrt{q}$ to 10^{-12} (Table 1, line (U)). $\square$

Remark 2.3 (What this says about the Riemann hypothesis). Kang–Li–Wang [2, Thm. 1.5.1] prove that Y is Ramanujan iff the nontrivial zeros of $\det P(u)$ have $|u| = q^{-1}$, iff the nontrivial zeros of $\det(I - uL_E)$ have $|u| \in \{q^{-1}, q^{-1/2}\}$. Theorem 2.2 shows that the second clause of the edge criterion is automatic: the zeros of $\det(I - uL_E)$ off the pencil are always on $|u| = q^{-1/2}$. The Riemann hypothesis for the zeta function of an $\tilde{A}_2$ complex is therefore exactly the Ramanujan property of the *vertex* spectrum, and the “parahoric” zeros are on the critical line for every complex, Ramanujan or not — for the same reason that the Hashimoto remainder of a graph is ± 1 : it is a local, not a global, statement.

3. THE FACTORIZATION THEOREM

Theorem 3.1 (Parahoric factorization). *For every $\tilde{A}_2$ complex of order q , with $K_Y(u) := \det(I - uC'|_{\ker \Psi})$,*

$$\begin{aligned} \det(I - uL_E) \cdot K_Y(u) &= \det P(u) \cdot G(u), \\ G(u) &= \det(I - uL_E|_W) \in \mathbb{Z}[u], \quad \deg G = \dim W = |E| - \text{rk } \Psi, \end{aligned}$$

K_Y divides $\det P$, and all roots of G lie on $|u| = q^{-1/2}$.

Proof. By Theorem 2.1, $L_E = L_E|_{\text{im } \Psi} \oplus R$, so $\det(I - uL_E) = \det(I - uL_E|_{\text{im } \Psi}) G(u)$. Ψ induces an isomorphism $\ell^2(V)^3 / \ker \Psi \rightarrow \text{im } \Psi$ intertwining the map induced by C' with $L_E|_{\text{im } \Psi}$, and $\ker \Psi$ is C' -invariant ($\Psi C'v = L_E \Psi v = 0$); hence $\det(I - uC') = K_Y(u) \det(I - uL_E|_{\text{im } \Psi})$. Finally $\det(I - uC') = \det P(u)$: the blocks of C' commute, and the formal determinant of $I - uC'$ expanded along the first block row is $(I - uA_1 + qu^2 A_2) + qu \cdot (-q^2 u^2) = P(u)$ [XI, Prop. 3.4]. Multiply out. Degree and roots: Theorem 2.2, R invertible. $\square$

Proposition 3.2 (The kernel of Ψ). *The Gram matrix*

$$\Psi^t \Psi = \begin{pmatrix} NI & A_1 & (q+1)A_2 \\ A_2 & NI & (q+1)A_1 \\ (q+1)A_1 & (q+1)A_2 & qNI + A_1 A_2 \end{pmatrix}$$

lies in M_3 of the commutative algebra $\mathbb{Q}[A_1, A_2]$; for a joint eigenvector with $A_1 f = \lambda f$, $A_2 f = \bar{\lambda} f$ it is the Hermitian matrix

$$H(\lambda) = \begin{pmatrix} N & \lambda & (q+1)\bar{\lambda} \\ \bar{\lambda} & N & (q+1)\lambda \\ (q+1)\lambda & (q+1)\bar{\lambda} & qN + |\lambda|^2 \end{pmatrix},$$

$$\det H(\lambda) = (qN + |\lambda|^2)(N^2 - |\lambda|^2) - 2N(q+1)^2|\lambda|^2 + 2(q+1)^2 \operatorname{Re} \lambda^3,$$

and $\dim \ker \Psi = \sum_{\lambda} \dim \ker H(\lambda)$ over the eigenvalues of A_1 with multiplicity. For the three trivial eigenvalues $\lambda = N\omega^j$, H has rank one, so $\dim \ker \Psi \geq 6$ [XI, Cor. 3.5(i)]. Say Y is non-degenerate (ND) if $\det H(\lambda) \neq 0$ for every nontrivial eigenvalue. Under (ND): $\ker \Psi$ is the 6-dimensional space of type-constant triples (f, g, h) with $f_i + g_{i+1} + (q+1)h_{i+2} = 0$,

$$K_Y(u) = (1 - u^3)(1 - q^3 u^3), \quad \deg G = 4n + 6,$$

and Theorem 3.1 is the identity of [XI, Cor. 3.5(ii)]. $Y_{21}, Y_{24}, Y_{42}, Y_{168}$ satisfy (ND); the thin torus T_6 ($q = 1$) does not (Table 1, line (K)).

Proof. The block entries are [XII, Thm. 3.1]; A_1 is normal, so $\ell^2(V)$ is an orthogonal sum of joint eigenlines and $\Psi^t \Psi$ is the direct sum of the $H(\lambda)$. Expanding $\det H$ with $x = \lambda$, $y = \bar{\lambda}$, $r = xy$, $c = q+1$, $D = qN + r$: $N(ND - c^2 r) - x(yD - c^2 x^2) + cy(cy^2 - cNx) = D(N^2 - r) - 2Nc^2 r + c^2(x^3 + y^3)$. For $\lambda = N$ the rows are (N, N, cN) , (N, N, cN) , $(cN, cN, (c-1)N + N^2)$, and $(c-1)N + N^2 - c^2 N = N(q + N - (q+1)^2) = 0$, so the rank is one; the ω -rotations are conjugate. Under (ND) the kernel is the 6-dimensional type-constant space described (it is C' -invariant and contained in the 9-dimensional space V_0 of type-constant triples). C' on V_0 has characteristic polynomial $\prod_j P_{N\omega^j}(u) = (1 - u^3)(1 - q^3 u^3)(1 - q^6 u^3)$, and on the quotient $V_0 / \ker \Psi \cong \Psi(V_0)$, the type-constant edge functions, L_E acts by q^2 times the type rotation, with characteristic polynomial $1 - q^6 u^3$; dividing gives K_Y . Then $\deg G = |E| - (3n - 6) = 4n + 6$. $\square$

4. THE CHAMBER COMPANION AND THE KANG-LI IDENTITY

A *pointed chamber* is a pair (e, z) with $e = (x \rightarrow y) \in E$ and $\{x, y, z\} \in F$; write it $(x, y; z)$. There are $|P| = 3|F| = (q+1)|E|$ of them. The Kang-Li operator [1] is

$$L_B((x, y; z), (y, z; w)) = 1 \quad \text{iff} \quad w \neq x,$$

row sums q ; it is the chamber flow L_F of [XI, Rem. 3.6] and [XII, Def. 4.1]. Let $\sigma, \rho, \tau: P \rightarrow E$ send $(x, y; z)$ to $(x \rightarrow y)$, $(y \rightarrow z)$, $(z \rightarrow x)$, and $\Phi := [\sigma^* \rho^* \tau^*]: \ell^2(E)^3 \rightarrow \ell^2(P)$, $\Phi(f, g, h)(x, y; z) = f(x, y) + g(y, z) + h(z, x)$.

Theorem 4.1 (Chamber companion).

$$L_B \Phi = \Phi B, \quad B = \begin{pmatrix} 0 & 0 & -I_E \\ qI_E & K & K^t \\ 0 & -I_E & 0 \end{pmatrix}, \quad \det(I + uB) = \det((1 + qu^3)I_E + uK + u^2 K^t).$$

Proof. All sums run over the q vertices $w \neq x$ with $\{y, z, w\} \in F$; with $K((y, z), (z, w)) = [\{y, z, w\} \in F]$,

$$(L_B \sigma^* f)(x, y; z) = \sum_w f(y, z) = q(\rho^* f)(x, y; z),$$

$$(L_B \rho^* g)(x, y; z) = \sum_w g(z, w) = (Kg)(y, z) - g(z, x) = (\rho^* Kg - \tau^* g)(x, y; z),$$

$$(L_B \tau^* h)(x, y; z) = \sum_w h(w, y) = (K^t h)(y, z) - h(x, y) = (\rho^* K^t h - \sigma^* h)(x, y; z).$$

Reading the three columns gives B . For the determinant, permute the block rows and columns of $I + uB$ to

$$\begin{pmatrix} I & -uI & 0 \\ 0 & I & -uI \\ quI & uK^t & I + uK \end{pmatrix};$$

the upper-left 2×2 block is unipotent with inverse $\begin{pmatrix} I & uI \\ 0 & I \end{pmatrix}$, and the Schur complement is

$$I + uK - (quI \quad uK^t) \begin{pmatrix} I & uI \\ 0 & I \end{pmatrix} \begin{pmatrix} 0 \\ -uI \end{pmatrix} = I + uK + qu^3I + u^2K^t.$$

No commutation of K, K^t is used. Exact on all five complexes (Table 1, line (B)). $\square$

Theorem 4.2 (Kang–Li identity in companion form). *For an $\tilde{A}_2$ complex of order q satisfying (ND),*

$$\det((1 + qu^3)I_E + uK + u^2K^t) = (1 - u^3)^n \frac{\det(I - uL_E) \det(I - u^2L_E)}{\det P(u)},$$

and the restriction of B to W^3 has characteristic polynomial $G(u)G(u^2)$.

Proof. $\ell^2(E)^3 = (\text{im } \Psi)^3 \oplus W^3$ orthogonally, both summands B -invariant (Theorem 2.1). On W , $K = T^tS - L_E$ acts as $-R$ and K^t as $-R^t$, so $B|_{W^3} = \begin{pmatrix} 0 & 0 & -I \\ qI & -R & -R^t \\ 0 & -I & 0 \end{pmatrix}$ and, by the elimination of Theorem 4.1, $\det(I + uB|_{W^3}) = \det((1 + qu^3)I - uR - u^2R^t) = \det((I - uR)(I - u^2R^t)) = G(u)G(u^2)$, using $RR^t = qI$ (Theorem 2.2) and $\det(I - u^2R^t) = G(u^2)$. On $(\text{im } \Psi)^3$, $\Psi^{\oplus 3}$ intertwines B with $\hat{B} = \begin{pmatrix} 0 & 0 & -I \\ qI & \hat{C} & \hat{C}' \\ 0 & -I & 0 \end{pmatrix}$ on $\ell^2(V)^9$ modulo $(\ker \Psi)^3$, so $\det(I + uB|_{(\text{im } \Psi)^3}) = \det(I + u\hat{B}) / \det(I + u\hat{B}|_{(\ker \Psi)^3})$.

The same elimination gives $\det(I + u\hat{B}) = \det((1 + qu^3)I + u\hat{C} + u^2\hat{C}')$, a 3×3 block matrix over $\mathbb{Q}[A_1, A_2]$; on a joint eigenline $(a, b) = (\lambda, \bar{\lambda})$ it is

$$D_{a,b}(u) = \det \begin{pmatrix} 1 + qu^3 & (q+1)u^2 & qu + au^2 \\ (q+1)u & 1 + qu^3 & bu + qu^2 \\ u^2 & u & 1 + qu^3 \end{pmatrix} = (1 - u^3)(1 - bu^2 + qau^4 - q^3u^6)$$

(a polynomial identity in a, b, q, u , verified by expansion), so $\det(I + u\hat{B}) = (1 - u^3)^n \det(I - u^2A_2 + qu^4A_1 - q^3u^6) = (1 - u^3)^n \det P(u^2)$, the last step by transposition. Under (ND), $(\ker \Psi)^3 \subset V_0^3$ with V_0 the type-constant space; $\det(I + u\hat{B}|_{V_0^3}) = \prod_j D_{N\omega^j, N\omega^{-j}}(u) = (1 - u^3)^3(1 - u^6)(1 - q^3u^6)(1 - q^6u^6)$, and on $\Psi(V_0)^3$ — type-constant edge functions, on which K and K^t act as $(q+1)$ times the type rotation and its inverse — the factor is $\prod_j (1 + qu^3 + (q+1)u\omega^j + (q+1)u^2\omega^{2j}) = \prod_j (1 + u\omega^j + u^2\omega^{2j})(1 + qu\omega^j) = (1 - u^3)^2(1 + q^3u^3)$; the quotient is $\det(I + u\hat{B}|_{(\ker \Psi)^3}) = (1 - u^3)(1 - q^3u^3)(1 - u^6)(1 - q^3u^6) = K_Y(u)K_Y(u^2)$. Assembling,

$$\det(I + uB) = (1 - u^3)^n \frac{\det P(u^2) G(u) G(u^2)}{K_Y(u) K_Y(u^2)},$$

and Theorem 3.1 at u and at u^2 turns the right side into the statement. Certified modulo two 31-bit primes at $3|E| + 1$ points on $Y_{21}, Y_{24}, Y_{42}, T_6$ (a polynomial identity of degree $\leq 3|E|$) and at random points on Y_{168} (Table 1, line (B)). $\square$

Corollary 4.3 (Kang–Li’s zeta identity). $\det(I + uL_B) K_\Phi(u) = \det(I + uB) G_\Phi(u)$ with K_Φ, G_Φ the characteristic polynomials of B on $\ker \Phi$ and of L_B on $\ell^2(P)/\text{im } \Phi$. For $q = 2$, where $|P| = 3|E|$, $K_\Phi = G_\Phi$ on the four thick complexes, and hence, with $\chi(Y) = n$,

$$\frac{(1 - u^3)^{\chi(Y)}}{\det P(u)} = \frac{\det(I + uL_B)}{\det(I - uL_E) \det(I - u^2L_E^t)},$$

which is [1, Thm. C], [2, eq. (1.6)], for the zeta function $Z(Y, u) = \det(I - uL_E)^{-1}$. For general q , [1] gives $G_\Phi/K_\Phi = (1 - u^3)^{\chi(Y)-n}$; verified on T_6 , where $\chi = 0$ and $n = 36$. The eigenvalues of L_B off the pencil are the roots of $G(u)G(u^2)$: moduli $q^{1/2}$ and $q^{1/4}$, the latter being square roots of the unitary remainder — the $q^{-1/4}$ zeros of [2, Thm. 1.5.1(4)].

Proof. Φ intertwines, $\ker \Phi$ is B -invariant and $\operatorname{im} \Phi$ is L_B -invariant, so both characteristic polynomials factor through $\ell^2(E)^3 / \ker \Phi \cong \operatorname{im} \Phi$. The rest is Theorem 4.2, $\det(I - u^2 L_E) = \det(I - u^2 L_E^t)$, and the computations of Table 1; the equality $K_\Phi = G_\Phi$ and the general ratio are not proved here and are the one point where [1] is used. $\square$

5. THE VERIFICATION BATTERY

check	complexes	result
(D) $L_E L_E^t = qI + (q^2 - q)T^t T$, $L_E^t L_E = qI + (q^2 - q)S^t S$; $L_E \Psi = \Psi C'$, $L_E^t \Psi = \Psi C''$, $K\Psi = \Psi \tilde{C}$, $K^t \Psi = \Psi \tilde{C}'$ adjacency version of [XI, Def. 3.1] equals the chamber version	all five Y_{168}, T_6 yes; Y_{21}, Y_{24}, Y_{42} no	exact exact
(K) $\operatorname{rk} \Psi = 3n - 6$ (mod two primes + explicit rational kernel); $K_Y = (1 - u^3)(1 - q^3 u^3)$; $\dim W = 4n + 6$ T_6 : $\operatorname{rk} \Psi = 93$, $\dim \ker \Psi = 15$, (ND) fails, $\dim W = 15$, K_Y of degree 15	$Y_{21}, Y_{24}, Y_{42}, Y_{168}$ T_6	exact exact
(U) $RR^t = R^t R = qI$ on W ; $ \operatorname{spec} R = \sqrt{q}$ to 10^{-12}	all five	numeric
(E) $\det(I - uL_E)K_Y = \det P \cdot G$ in $\mathbb{Z}[u]$; $\deg G = \dim W$; functional equation; roots on $ u = q^{-1/2}$ Y_{168} : six missing pencil roots $\{1, \omega, \omega^2, q, q\omega, q\omega^2\}$, 678 remainder eigenvalues of modulus $\sqrt{q}$	$Y_{21}, Y_{24}, Y_{42}, T_6$ Y_{168}	exact (roots numeric) numeric
(B) $L_B \Phi = \Phi B$; Theorem 4.2 at $3 E + 1$ points mod two primes; $\det(I + uL_B) = \det(I + uB)$ ($q = 2$); ratio $(1 - u^3)^{\chi-n}$ on T_6	all five (Y_{168} : random points)	exact / modular
(V) nontrivial pencil roots of modulus q (vertex Ramanujan property)	all five	numeric

TABLE 1. Verification (pgl3_parahoric.py). Complexes: $Y_{21}, Y_{24}, Y_{42}, Y_{168}$ ($q = 2$) and T_6 ($q = 1$) of [X].

6. SCOPE, RESIDUE, CORRECTIONS

Remark 6.1 (Solved, open, impossible). *Solved*: [XI, Rem. 5.1(b)] in full — the determinant identity for all $\tilde{A}_2$ complexes (Theorem 3.1, with the exact form of the trivial factor under (ND), Proposition 3.2) and the identification of G with Kang–Li’s chamber factor (Theorem 4.2: $G(u)G(u^2)$ is the parahoric block of the chamber companion); the Riemann hypothesis for the parahoric factor, unconditionally (Theorem 2.2); the Kang–Li identity itself in companion form, with a proof by the degeneracy calculus of [XII] (Theorem 4.2). *Open*: (a) a direct proof that $G_\Phi/K_\Phi = (1 - u^3)^{\chi-n}$ (Corollary 4.3), which would make the Kang–Li identity independent of [1]; (b) an exact (rather than numeric) certificate of the vertex Ramanujan property of $Y_{21}, \dots, Y_{168}$, i.e. of the Riemann hypothesis for their zeta functions; (c) the exact polynomial identity on Y_{168} , verified here only structurally and at random points modulo primes; (d) the $\tilde{A}_2 \rightarrow \tilde{A}_d$ generalization, where the parahoric remainders should again be $\sqrt{q^k}$ times orthogonal matrices. *Impossible or false as stated*: see Remark 6.2.

Remark 6.2 (Corrections to the task specification). (1) The identity $\det(I - uL_E)(1 - u^3)(1 - q^3 u^3) = \det P(u) \det(I - uL_F^{\operatorname{par}})$ with L_F^{par} an operator on the $3|F|$ pointed chambers is false as stated: the right factor would have degree up to $3|F| = (q+1)|E|$, not $4n + 6$. The remainder G is the characteristic polynomial of L_E on $W \subset \ell^2(E)$, and it enters the chamber operator as the block $G(u)G(u^2)$. (2) “ $L_F^{\operatorname{par}} \cong \operatorname{im}(\Psi)^\perp$ ” is false: $\dim W = 4n + 6$ while L_B acts on $(q+1)|E|$ coordinates; the correct relation is $L_B \Phi = \Phi B$ with B on $\ell^2(E)^3$. (3) The “Riemann hypothesis for $\tilde{A}_2$ complexes” for the parahoric poles is true but needs no Ramanujan hypothesis: it holds for every complex (Theorem 2.2); the Ramanujan property is exactly the vertex condition (Remark 2.3). (4) The formula requires (ND); on T_6 the trivial factor has degree 15 and $\deg G = 15 \neq 4n + 6$. (5) The “block triangularization” is in fact a block diagonalization: $\operatorname{im} \Psi$ is invariant under L_E^t too. (6) Our own [XI, Def. 3.1] is corrected (Remark 1.2). (7) The representation-theoretic reading of the remainder as “the subdominant Iwahori–Hecke spectrum on a parahoric” is [2]’s; here the remainder

is characterized combinatorially and its unitarity proved, which is more than the representation theory gives without a temperedness input.

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Unrefereed preprint. All numbered statements are proved from the link structure of an $\tilde{A}_2$ complex and linear algebra; the one external input is the relation between L_B and its companion for $q \neq 2$ in Corollary 4.3, quoted from [1]. The modular certificates of Theorem 4.2 are identities in $\mathbb{F}_p[u]$ for two primes, not in $\mathbb{Z}[u]$; the theorem itself is proved. Bibliographic data of [1, 2, 3] were checked against publisher pages.

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